GINE versus Ridge: full-budget nested evaluation

Matched paired dataset: 318 scans / 252 participants; CN evaluation: 237 scans / 189 participants.

Five participant-grouped outer folds, three inner folds; early stopping separated inside inner training.

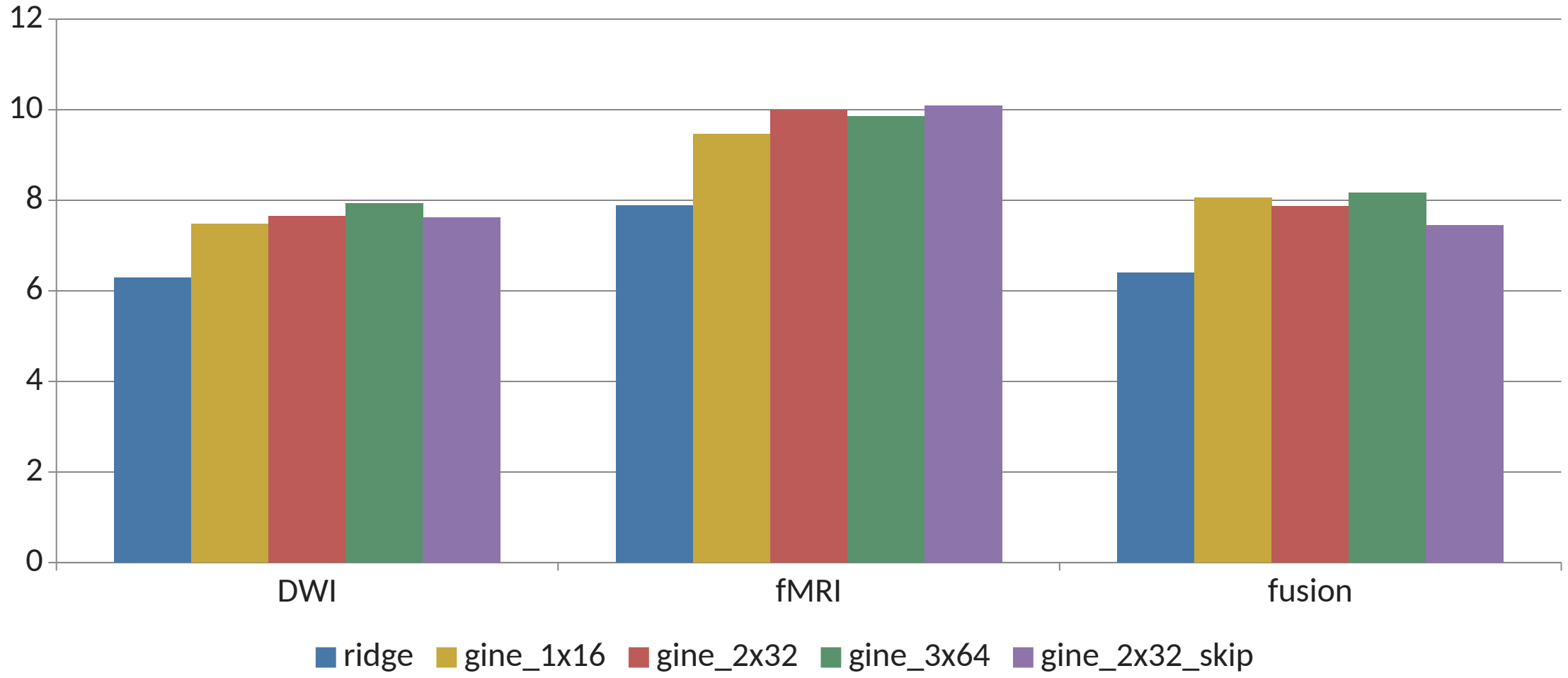
GINE: 1×16, 2×32, 3×64 and 2×32 residual. Two sparsity levels selected inside inner CV.

Training-only mean-only ComBat versus no harmonization; imaging-only predictors.

Training-size study: prespecified 2×32 versus Ridge. Unsupported fractions are disclosed.

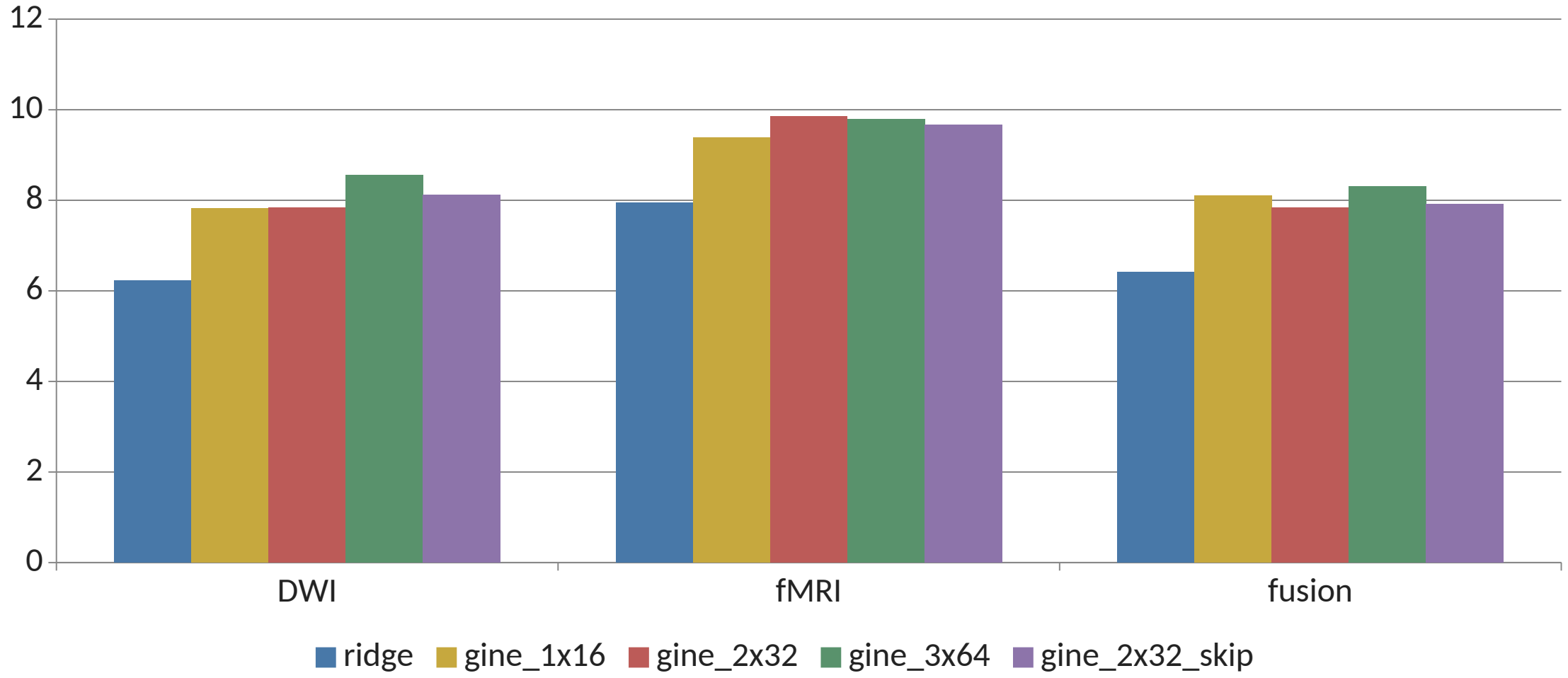
Development reanalysis; clinical dates, acquisition protocols and ROI ordering remain unverified. Completed audited arms only.

OOF MAE — none



Native editable chart. Point estimates; see HTML/CSV for participant-bootstrap intervals and paired differences. No architecture is selected using outer-test scores.

OOF MAE — age_mean



Native editable chart. Point estimates; see HTML/CSV for participant-bootstrap intervals and paired differences. No architecture is selected using outer-test scores.

Training and validation diagnostics — age_mean

arm	modality	parameters	selected epoch median	mean training MAE	mean stopping MAE	mean stopping train gap
gine_1x16	DWI	1977	74.00	4.84	8.32	3.49
gine_1x16	fMRI	1977	53.00	7.14	10.35	3.21
gine_1x16	fusion	3657	64.00	4.44	8.55	4.11
gine_2x32	DWI	6793	30.00	4.94	8.70	3.76
gine_2x32	fMRI	6793	46.00	5.46	9.89	4.43
gine_2x32	fusion	9849	37.00	3.73	8.55	4.82
gine_2x32_skip	DWI	10505	21.00	5.16	8.58	3.42
gine_2x32_skip	fMRI	10505	32.00	5.40	10.07	4.67
gine_2x32_skip	fusion	16249	35.00	3.28	8.03	4.74
gine_3x64	DWI	29161	18.00	4.06	8.17	4.11
gine_3x64	fMRI	29161	27.00	4.15	9.53	5.38
gine_3x64	fusion	35001	41.00	3.16	8.21	5.05

Mean across selected-candidate inner stopping splits. These validation errors are not outer-test estimates; each run report contains epoch-by-epoch curves.